

PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 22

Facts

- Like a random variable is probabilistically specified by its distribution, a stochastic process is specified by its finite dimensional distributions, *i.e.*, by

$$P(X_0 = i_0, X_1 = i_1, \dots, X_n = i_n)$$

for $n \geq 0$ and all $i_0, i_1, \dots, i_n \in S$, where S is the state space.

Facts

- Let $P(X_0 = i) = \mu_i$ for $i \in S$. Then

$$P(X_0 = i_0, X_1 = i_1, \dots, X_n = i_n) = \left(\prod_{k=0}^{n-1} p_{i_k i_{k+1}} \right) \mu_{i_0}.$$

Thus, a MC is probabilistically specified by its initial distribution and one-step TPM.

Markov Property

Theorem: Let $\{X_n : n \geq 0\}$ be a MC with initial distribution μ and TPM P . Then conditional on $X_m = i$, $\{X_{m+n}\}_{n \geq 0}$ is a MC with initial distribution δ_i and TPM P .

n -step Transition Probability

- n -step transition probability is defined by

$$p_{ij}^{(n)} = P(X_n = j | X_0 = i).$$

- Note that

$$p_{ij}^{(n)} = P(X_n = j | X_0 = i) = P(X_{n+k} = j | X_k = i).$$

Chapman-Kolmogorov Equations

Theorem: Consider a MC having state space $S = \{0, 1, \dots\}$ and one-step transition probabilities p_{ij} for $i, j \in S$. The Chapman-Kolmogorov equations are given by

$$p_{ij}^{(m+n)} = \sum_{k=0}^{\infty} p_{ik}^{(m)} p_{kj}^{(n)}$$

for all $m, n \geq 0$ and all $i, j = 0, 1, \dots$

n -step TPM

- The matrix $\left(p_{ij}^{(n)}\right)_{i,j \geq 0}$ is called n -step TPM.
- The matrix form of Chapman-Kolmogorov equations is

$$P^{(n+m)} = P^{(n)} P^{(m)}.$$

- Note that $P^{(1)} = P$.
- Writing $n = (n-1) + 1$, we have $P^{(n)} = P^n$.

Example

Example 1: Suppose that the chance of rain tomorrow depends on previous weather conditions only through whether or not it is raining today and not on past weather condition. Suppose that if it is raining today, then it will rain tomorrow with probability $\alpha = 0.7$. If it is not raining today, then it will rain tomorrow with probability $\beta = 0.4$. Calculate the probability that it will rain four days from today given that it is raining today.

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- 0: the state that it is raining.
- 1: the state that is not raining.
- X_n : state of the n th day.
- $\{X_n\}_{n \geq 0}$ is a MC with state space $\{0, 1\}$.
- We want to find $p_{00}^{(4)}$.

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- $P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$

- $P^{(4)} = \begin{pmatrix} 0.5749 & 0.4241 \\ 0.5668 & 0.4332 \end{pmatrix}$

- $p_{00}^{(4)} = 0.5749$

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- X_n : Number of occupied urns after distribution of n balls
- $\{X_n\}_{n \geq 0}$ is a MC with state space $S = \{0, 1, \dots, 8\}$
- $p_{i,i} = \frac{i}{8}$ and $p_{i,i+1} = \frac{8-i}{8}$ for $i = 0, 1, \dots, 8$
- We want to find $p_{03}^{(9)}$
- $p_{03}^{(9)} = \sum_{k=0}^8 p_{0k} p_{k3}^{(8)} = p_{13}^{(8)}$ as $p_{01} = 1$
- Starting with 1 occupied urn, we need to find the probability of exactly 3 occupied urns after distribution 8 balls
- It is difficult to calculate P^9 .

Example

Example 2: Suppose that balls are successively distributed among 8 urns, with each ball being equally likely to be put in any of these urns. What is the probability that there will be exactly 3 occupied urns after 9 balls have been distributed?

- We can simplify the problem simple by defining the MC cleverly
- X_n does not decrease
- Collapse all states 4, 5, $\dots$, 8 into one
- The states of the new MC $\{Y_n\}_{n \geq 0}$ are
 - i : i urns are occupied, $i = 1, 2, 3$
 - 4: at least four urns are occupied
- We want to find $q_{13}^{(8)} = P(Y_8 = 3 | Y_0 = 1)$.

Example

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- TPM of $\{Y_n\}$ is $Q = \begin{pmatrix} 1/8 & 7/8 & 0 & 0 \\ 0 & 2/8 & 6/8 & 0 \\ 0 & 0 & 3/8 & 5/8 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

- $Q^4 = \begin{pmatrix} 0.0002 & 0.0256 & 0.2563 & 0.7178 \\ 0 & 0.0039 & 0.0952 & 0.9009 \\ 0 & 0 & 0.0198 & 0.9802 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

- $q_{13}^{(8)} = 0.0002 \times 0.2563 + 0.0256 \times 0.0952 + 0.2563 \times 0.0198 = 0.00756$

Example

Example 3: In a sequence of independent flips of a fair coin, let N denote the number of flips until there is a run of three consecutive heads. Find $P(N \leq 8)$ and $P(N = 8)$.

Example

Example 4: In a sequence of independent flips of a fair coin, let N denote the number of flips until there is a run of three consecutive heads. Find $P(N \leq 8)$ and $P(N = 8)$.

- Consider a MC $\{X_n\}$ with following states:
 - i : Currently on a run of i consecutive heads, $i = 0, 1, 2$
 - 3: A run of three consecutive heads has already occurred

- TPM of the MC is $P = \begin{pmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0.5 & 0 & 0 & 0.5 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

- $N \leq 8$ if and only if $X_8 = 3$.
- Required probability is $P(N \leq 8) = p_{03}^{(8)} = \frac{107}{256}$
- $P(N = 8) = P(N \leq 8) - P(N \leq 7) = p_{03}^{(8)} - p_{03}^{(7)}$